**Learning Objective 3.2:** I can derive the array factor for an arbitrary linear array and apply pattern multiplication.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Reading a sweep.** A lab group sweeps the PHASER with all eight elements active and a source at boresight. Working at $10.3$ GHz with element spacing $d = 14$ mm, they record the trace below. The sweep step is $2.8125^\circ$ and the peak reads $-4.1$ dBFS.

| $\theta$ (°) | $0.0$ | $2.8$ | $5.6$ | $8.4$ | $11.3$ | $14.1$ | $16.9$ | $19.7$ | $22.5$ | $25.3$ |
|---|---|---|---|---|---|---|---|---|---|---|
| Power (dBFS) | $-4.1$ | $-4.6$ | $-6.2$ | $-9.1$ | $-14.3$ | $-24.1$ | $-22.0$ | $-17.4$ | $-16.6$ | $-18.1$ |

   (a) Using linear interpolation between the two samples that bracket the half-power level, find the half-power beamwidth. The trace is symmetric about $0^\circ$.

   $\theta_{\text{HP}}$ = ____

   (b) Find the first-null beamwidth as the sweep reports it, and compare it with FNBW $= 2 \arcsin(\lambda / N d)$.

   $\text{FNBW}_{\text{meas}}$ = ____

   $\text{FNBW}_{\text{calc}}$ = ____

   (c) Find the first sidelobe level relative to the peak, in dBc.

   SLL = ____

   (d) No sample in the sweep reads below about $-24$ dBFS, which is $20$ dB below the peak, yet the noise floor on this kit is $23$ dB below the peak. Explain in one or two sentences why the deepest sample sits $3$ dB above the floor rather than on it.

2. **Predict before you measure.** The same array is reconfigured by setting some element gains to zero. The spacing $d = 14$ mm and the frequency $10.3$ GHz do not change, so $\lambda = 29.1$ mm throughout.

   (a) E1, E2, E7, and E8 are set to zero, leaving E3 through E6 active — four adjacent elements. Predict the half-power beamwidth from $\theta_{\text{HP}} \approx 0.886\lambda/(Nd)$.

   $\theta_{\text{HP}}$ = ____

   (b) Now only E3, E4, and E5 are left on — three adjacent elements. Predict the half-power beamwidth and the first-null beamwidth.

   $\theta_{\text{HP}}$ = ____

   FNBW = ____

   (c) By how many dB should the peak of the three-element sweep sit below the peak of the full eight-element sweep, if every element gain that is still on remains at 100%?

   $\Delta$ = ____

   (d) A student proposes reaching the same $35^\circ$ beamwidth by leaving all eight elements on and turning every gain down to $37.5\%$. Explain in one or two sentences why that does not work.

3. **Reconciling a sidelobe.** A second group measures the first sidelobe of the full eight-element array at $-11.5$ dBc, while the array factor for $N = 8$ predicts $-12.8$ dB. Their noise floor sits at $-23$ dBc and the sweep step is $2.8125°$.

   (a) The floor adds noise power to the sidelobe power the receiver reports. Treating the two as uncorrelated, compute the level the sweep should report for a true $-12.8$ dB sidelobe sitting on a $-23$ dBc floor.

   SLL = ____

   (b) Their peak reading is also affected. The peak is $23$ dB above the floor by definition. By how much does the floor lift the peak reading, and which way does that push the reported sidelobe level in dBc?

   peak lift = ____

   (c) The floor explains $0.4$ dB of the $1.3$ dB discrepancy. Name two other contributions that plausibly make up the rest, and say for each whether it would move the reading up or down.

4. **The two-element case.** With only E4 and E5 active the array has two elements at $d = 14$ mm, and at $10.3$ GHz the array factor is $AF = \cos(\psi/2)$ with $\psi = kd\sin(\theta)$.

   (a) Find the half-power beamwidth exactly, without using the $0.886\lambda/(Nd)$ rule.

   $\theta_{\text{HP}}$ =

   (b) Show that the $0.886\lambda/(Nd)$ rule fails here, and state the percentage error it makes.

   error =

   (c) Evaluate $\lambda/2d$ and use it to explain why the lab table lists the first-null beamwidth of this array as $180^\circ$.

   $\lambda/2d$ =

   (d) How far below its own peak does this two-element pattern fall at $\theta = 90^\circ$? On one kit whose HB100 ran at $10.408$ GHz, the two-element trace fell about $27$ dB from its peak toward the edges of the scan. Repeat the calculation at that frequency and say what the comparison tells you about the edges of that trace.

   drop at $90^\circ$ =

   at $10.408$ GHz =

5. **Design judgment.** Your section is asked to demonstrate a $20^\circ$ beam on the PHASER at $10.3$ GHz, using only the Element Gains controls.

   (a) How many adjacent elements must be active to give a half-power beamwidth closest to $20^\circ$? Give the aperture and the resulting beamwidth.

   $N$ = ____

   $\theta_{\text{HP}}$ = ____

   (b) By how many dB does the voltage sum put this aperture's peak below the full eight-element peak? Can the sweep still read the first sidelobe, if the nulls fill to about $-23$ dBc of the trace's own peak?

   $\Delta$ = ____

   SLL = ____

   (c) A classmate suggests instead applying a Hann taper to all eight elements, since that also widens the beam to about $19.5^\circ$. State one advantage and one disadvantage of the taper against switching elements off, and say which you would choose for a demonstration whose purpose is to show that beamwidth depends on aperture.

**Documentation:** _____